

COMSOL Validation of GPU-Accelerated MRT-LBM Simulation of Convective Flows in Porous Media

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Abstract

This project validates the natural convection results from Molla et al. (2018), which utilizes a GPU-accelerated Multiple-Relaxation-Time Lattice Boltzmann Method (MRT-LBM) for simulating natural and mixed convection in porous media. Using COMSOL Multiphysics, the study independently reproduced key quantitative and qualitative results including average Nusselt numbers for both pure-fluid and porous media cases, across Rayleigh numbers $Ra = 10^3$ – 10^6 and Darcy numbers $Da = 10^{-2}$ – 10^{-7} . Qualitative validation through streamline, isotherm, velocity distribution, local Nusselt number comparison with Molla et al. (2018) confirmed correct flow physics across the Rayleigh number range. These results confirm the validity of the finite element simulation setup and provide independent support for the numerical methodology of the reference paper.

1. Introduction

Natural and mixed convection in fluid-saturated porous enclosures is a topic of significant engineering interest, with applications spanning geothermal energy systems, electronic cooling, thermal insulation, and oil reservoir management. Accurate numerical simulation of these flows requires methods that are both computationally stable over wide parameter ranges and efficient for large-scale computations.

Molla et al. (2018) addressed these requirements by developing a double MRT-LBM framework accelerated on an NVIDIA Tesla K40 GPU using CUDA C. Their model incorporated the Brinkman–Forchheimer extended Darcy model for porous media and used the D2Q9 lattice for fluid flow and the D2Q5 lattice for thermal transport. The GPU implementation achieved a maximum speedup of $144\times$ over a single-core CPU for a 2048×2048 lattice, demonstrating the computational efficiency of the approach for high Rayleigh number convection problems.

The objective of this project is to independently replicate the key quantitative and qualitative results of Molla et al. (2018) using COMSOL Multiphysics. Specifically, the simulations target the sections of the reference paper listed in Table 1.

Table 1: Sections of Molla et al. (2018) and corresponding COMSOL validation targets.

Section	Title in Molla et al. (2018)	COMSOL Validation
5.1	Lid-driven cavity flow for $Re = 10,000$ and $20,000$	—
5.2	Natural convection in square cavity for $Ra = 10^3$ to 10^9	Validated ($Ra = 10^3$ to 10^6)
5.3	Natural convection with porous media for $Ra = 10^3$ to 10^{10}	Validated
5.4	Mixed convection in a lid-driven square cavity	—
5.5	Mixed convection in lid-driven square cavity with porous media	—

2. Problem Description and Governing Equations

2.1 Physical Configuration

The physical configuration considered in this study is a two-dimensional square cavity of height H and width L (with $L = H$), filled with a fluid-saturated porous medium. As shown in Figure 2(A) of Molla et al. (2018), for the natural convection case the left wall is maintained at a higher temperature $T_h = 1$ and the right wall at a lower temperature $T_c = 0$, while the top and bottom walls are adiabatic (thermally insulated). No-slip velocity boundary conditions are imposed on all four walls.

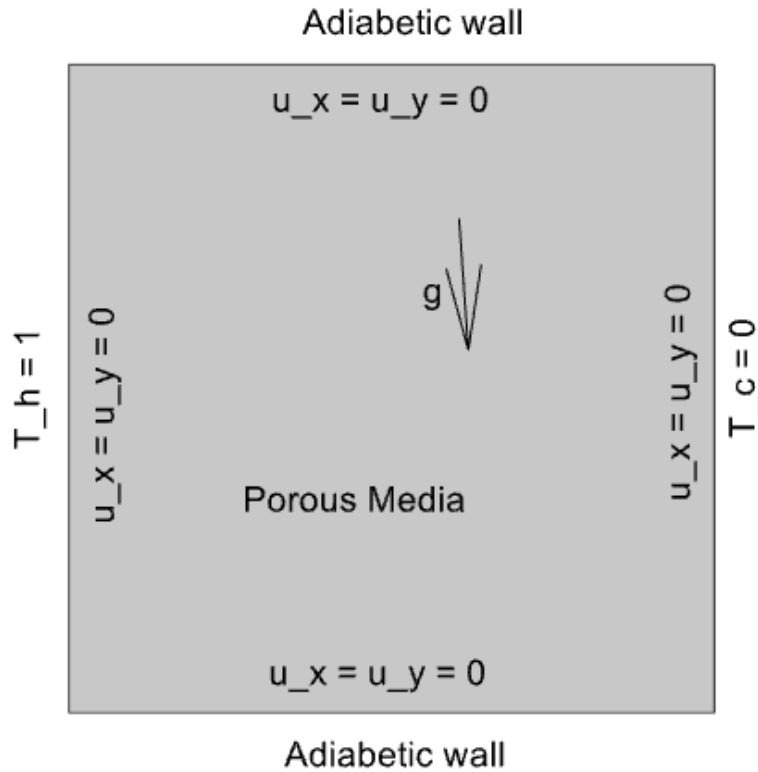


Figure 1: Schematic of the 2D square cavity for natural convection.

2.2 Non-Dimensional Governing Equations

The non-dimensional governing equations for laminar, incompressible, 2D natural convection flow through porous media are given as follows [1, 3]:

Continuity Equation:

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0 \quad (1)$$

x-Momentum Equation:

$$\frac{\partial u_x}{\partial t} + \frac{u_x}{\varepsilon} \frac{\partial u_x}{\partial x} + \frac{u_y}{\varepsilon} \frac{\partial u_x}{\partial y} = -\varepsilon \frac{\partial p}{\partial x} + A \left(\frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial y^2} \right) + D \quad (2)$$

y-Momentum Equation:

$$\frac{\partial u_y}{\partial t} + \frac{u_x}{\varepsilon} \frac{\partial u_y}{\partial x} + \frac{u_y}{\varepsilon} \frac{\partial u_y}{\partial y} = -\varepsilon \frac{\partial p}{\partial y} + A \left(\frac{\partial^2 u_y}{\partial x^2} + \frac{\partial^2 u_y}{\partial y^2} \right) + B\varepsilon\theta + D \quad (3)$$

Energy Equation:

$$\frac{\partial \theta}{\partial t} + u_x \frac{\partial \theta}{\partial x} + u_y \frac{\partial \theta}{\partial y} = C \left(\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} \right) \quad (4)$$

where u_x and u_y are the velocity components in the x and y directions, p is the non-dimensional pressure, $\theta = (T - T_c)/(T_h - T_c)$ is the non-dimensional temperature, and ε is the porosity of the porous medium.

For the natural convection case considered in this validation: $A = \text{Pr}$ (viscous diffusion term), $B = \text{Ra} \cdot \text{Pr}$ (buoyancy term), and $C = 1$ (thermal diffusion term), where Pr is the Prandtl number and Ra is the Rayleigh number. The term D is the porous media body force defined in Section 2.3.

2.3 Porous Media Body Force

The body force term arising from the presence of the porous medium is defined through the volume force vector $\mathbf{F}$ [1]:

$$\mathbf{F} = D + \varepsilon G \quad (5)$$

$$\mathbf{F} = -\frac{\varepsilon \nu}{K} \mathbf{u} - \frac{1.75}{\sqrt{150 \varepsilon K}} |\mathbf{u}| \mathbf{u} + \varepsilon G \quad (6)$$

where $K = \text{Da} \cdot H^2$ is the permeability of the porous medium, Da is the Darcy number, H is the height of the cavity, and $G = g\beta(T - T_m)$ is the buoyancy term, with $T_m = (T_h + T_c)/2$ being the mean reference temperature, g the gravitational acceleration, and β the coefficient of thermal expansion.

2.4 Average Nusselt Number

The primary quantity of interest for heat transfer assessment is the average Nusselt number $\overline{\text{Nu}}$. For the natural convection case, the local and average Nusselt numbers on the hot left wall are

defined as:

$$\text{Nu}(y) = - \left. \frac{\partial \theta}{\partial x} \right|_{\text{wall}} \quad (7)$$

$$\overline{\text{Nu}} = \frac{1}{H} \int_0^H \text{Nu}(y) dy \quad (8)$$

In the COMSOL Multiphysics simulations, the average Nusselt number is evaluated by computing the line-averaged negative temperature gradient normal to the hot wall, which in non-dimensional form with $\Delta T = 1$ and $H = 1$ reduces to the line average of $-\partial T / \partial x$ along the left boundary.

3. Results and Validation

3.1 Pure-Fluid Natural Convection

3.1.1 Qualitative Validation: Streamlines and Isotherms

Streamlines and isotherms from the present COMSOL simulation are compared against the benchmark results of De Vahl Davis (1983) for $\text{Ra} = 10^4$ and 10^6 . The benchmark contours and corresponding COMSOL results are shown side-by-side in Figures 2 and 3 for direct visual comparison.

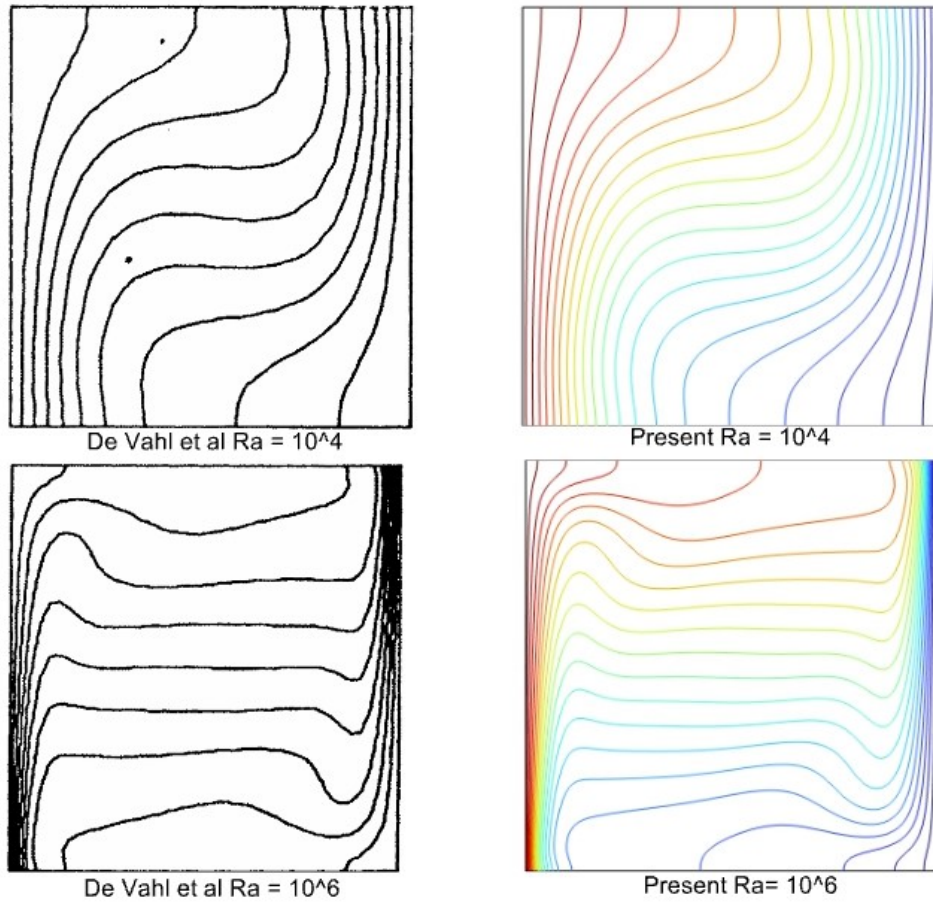


Figure 2: Isotherms at $\text{Ra} = 10^4$: (left) De Vahl Davis (1983), (right) present COMSOL.

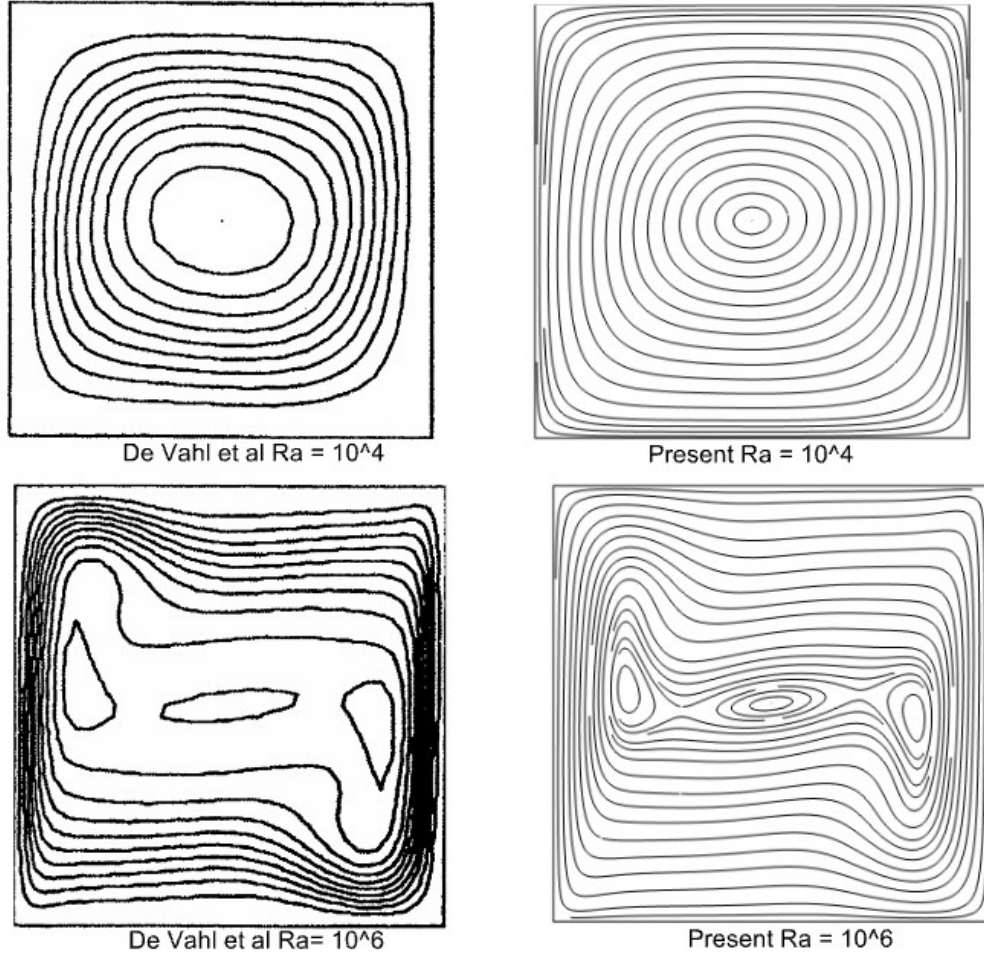


Figure 3: Streamlines at $Ra = 10^6$: (left) De Vahl Davis (1983), (right) present COMSOL.

3.1.2 Quantitative Validation: Average Nusselt Number

Table 2 compares the average Nusselt number obtained from the present COMSOL simulation with the finite difference benchmark of De Vahl Davis (1983) and the MRT-LBM of Molla et al. (2018). For $Ra \geq 10^7$, the COMSOL stationary solver failed to converge, as discussed in Section 4, so no COMSOL values are reported for those cases.

Table 2: Comparison of average Nusselt number $\overline{Nu}$ for pure-fluid natural convection. Error(%) is computed relative to Molla et al. and De Vahl Davis.

Ra	de Vahl Davis 1983	Molla (2018)	Present	Error (%) vs Molla	Error (%) vs De Vahl
10^3	1.116	1.1299	1.1178	1.070891229	0.161290323
10^4	2.238	2.2655	2.2568	0.384021187	0.840035746
10^5	4.509	4.5106	4.6004	1.99086596	2.027056997
10^6	8.817	8.8135	8.9765	1.849435525	1.809005331
Ra	Vishnampet (2011)	Molla (2018)	Present		
10^7	16.81	16.609	Not converged	—	—
10^8	30.81	30.221	—	—	—
10^9	55.80	54.561	—	—	—

3.2 Porous Media Natural Convection

3.2.1 Qualitative Validation: Streamlines, Isotherms, Velocity Distributions, and Local Nusselt Number

Qualitative validation of the porous media case was carried out by comparing COMSOL results against Figure 8, Figure 10, and Figure 11 of Molla et al. (2018).

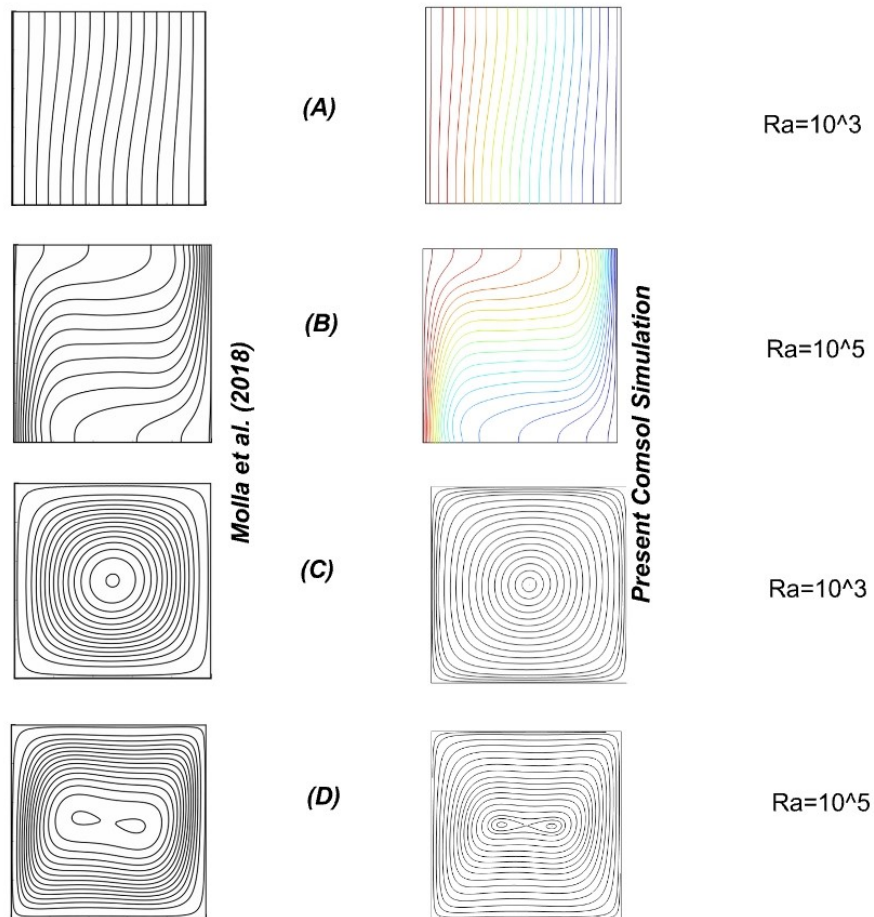


Figure 4: Comparison of streamlines and isotherms for porous media case corresponding to Figure 8 of Molla et al. (2018).

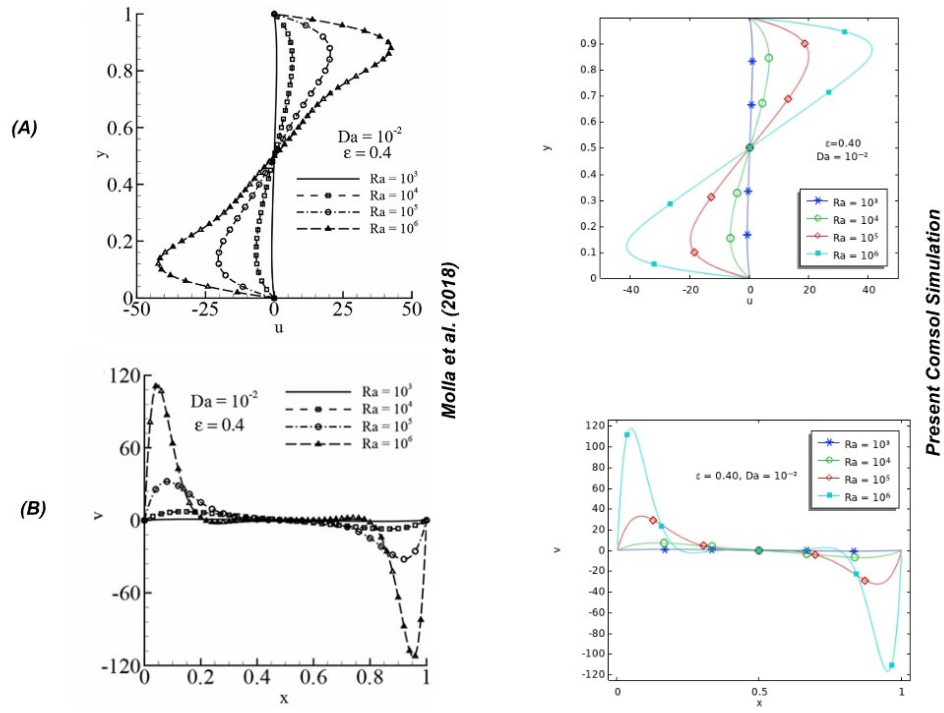


Figure 5: Comparison of velocity distributions corresponding to Figure 10 of Molla et al. (2018).

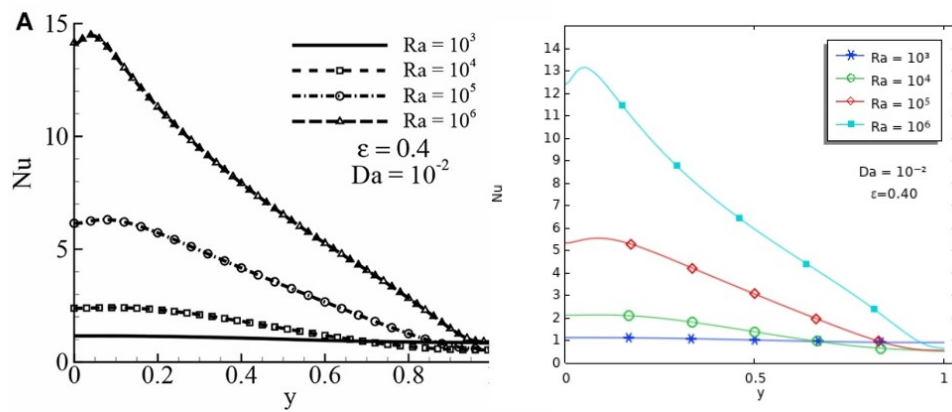


Figure 6: Comparison of local Nusselt number corresponding to Figure 11 of Molla et al. (2018).

3.2.2 Quantitative Validation: Average Nusselt Number

Table 3 reproduces the data from Table 4 of Molla et al. (2018) and compares it with our COMSOL results, capturing the average Nusselt numbers for the porous media model ($Pr = 1.0$) across a wide range of Ra , Da , and porosity values.

Table 3: Average Nusselt number $\overline{Nu}$ for porous media natural convection ($Pr = 1.0$). Comparison between Molla et al. (2018) and Present (COMSOL).

Da	Ra	$\varepsilon = 0.4$		$\varepsilon = 0.6$	
		Molla (2018)	Present	Molla (2018)	Present
10^{-2}	10^3	1.0197	1.0078	1.0240	1.0121
	10^4	1.3546	1.3644	1.5079	1.4949
	10^5	3.0293	3.0651	3.4855	3.4807
	10^6	6.1853	6.3785	7.1410	7.2258
10^{-4}	10^5	1.0681	1.0643	1.0914	1.0672
	10^6	2.6263	2.6000	2.7418	2.7086
	10^7	7.7831	7.8112	8.1243	8.5949
	10^8	16.8410	16.9760	19.2100	19.1810
10^{-6}	10^7	1.1095	1.0801	1.1100	1.0817
	10^8	2.9603	3.0714	3.1700	3.1034
	10^9	11.6730	12.0390	13.0500	12.5100
10^{-7}	10^8	1.1590	1.1040	1.1727	1.1159
	10^9	5.0369	3.3271	5.0402	3.4267
	10^{10}	12.6230	13.1380	13.1110	13.3710

4. Results and Discussion

4.1 Solver Convergence Limitation

For $Ra \geq 10^7$, the COMSOL stationary solver failed to converge. This is a known limitation of steady-state solvers applied to high Rayleigh number natural convection. Beyond $Ra \approx 10^6 - 10^7$, the flow transitions from steady laminar to time-periodic and eventually turbulent, meaning no steady-state solution exists. The MRT-LBM of Molla et al. (2018), being an explicit time-marching method, naturally handles this regime. Therefore, the present COMSOL validation is limited to $Ra \leq 10^6$.

4.2 Natural Convection Flow with Pure Fluid

The COMSOL results are compared with the reference paper, and they match well overall. The difference is around 5–10%, which is acceptable.

For the clear fluid case, the Nusselt number increases as the Rayleigh number increases. This indicates stronger heat transfer, which is physically correct. The streamlines and isotherms also show similar patterns to the reference results. Small differences occur because COMSOL and the reference study use different numerical methods.

4.3 Natural Convection Flow with Porous Media

For the porous media case, the trends are also correctly captured. As the Rayleigh number increases, heat transfer increases. As the Darcy number decreases, the flow becomes weaker and heat transfer reduces. Increasing porosity leads to higher heat transfer. All these behaviours match the reference results.

The small differences arise from how COMSOL and the MRT-LBM model handle the porous media terms. COMSOL uses the finite element method (FEM) to solve the governing equations, while the MRT-LBM uses the lattice Boltzmann method based on particle distribution functions and collision–streaming processes.

References

References

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